

Should I Dine Out in LA Today?

A Dashboard to Plan Your Dining Experience

The Problem

Most dining apps tell you where to eat, not whether it is a good day to go out at all!

Weather

Rain, extreme heat, and high winds discourage dining, but restaurant apps ignore these conditions.

Air Quality

Measuring AQI can be especially important for those with heart conditions/asthma or during fire season. Most apps ignore this factor.

Traffic

Traffic conditions can turn a 10-minute drive into 45 minutes. Knowing real time traffic conditions around the area helps determine where to go.

Restaurant Quality

Sometimes, we don't want to read 10 different reviews. ML scoring and Google Ratings help identify which restaurants are worth the trip.

The Solution

Weather and Air Quality



Collected by Open Weather Map | Temperature, Humidity, Wind Speed, AQI, etc.

Traffic



Collected by TomTom Traffic | Congestion Percentage (Current vs Free Flow)

Restaurant Quality

Collected by Google Places API | Rating, Review Count, Price Level, Cuisine, etc.



The Supabase Icon Means the data was collected then stored in a PostgreSQL Database hosted by Supabase.

The data is updated every hour via a GitHub Actions cronjob and data older than seven days is removed by a Supabase scheduled function.

The Model - Preprocessing Development

Dataset Characteristics

3,343 Training Restaurants

4 Training Cities (SD, SF, NY, Chicago)

Tested Using Cross Validation

- XGBoost
- Random Forest
- Logistic Regression

Deployed on Los Angeles Restaurant Data

Variables

Dependent Variable: Restaurant Quality Tiers

- Skip It, Decent, Worth It, Must Go
- Based on Percentiles of Ratings of Training Set

Independent Variable: Restaurant Features

- Review Count (log transformed b/c skew)
- Price Level (Google's 1-4 price tier)
- Cuisine Type (transformed and truncated)
- Has Hours (acts as a proxy for legitimacy)
- etc.

The Model - Evaluation

Important Notes

Labels are Derived from rating percentiles, which are a near-linear boundary.

The dependent variable directly utilizes the rating from the training data, which explains why logistic regression performs so well.

XGBoost performance is not terrible but still doesn't beat the performance of Logistic Regression.

All classes score F1 0.97-0.99, implying no adjacent-tier confusion (knowing the difference between 0 vs 1 and 0 vs 3).



Architecture

App Functionality

Gives Current Conditions on temperature, traffic for local neighborhoods, and weighted conditions scores.

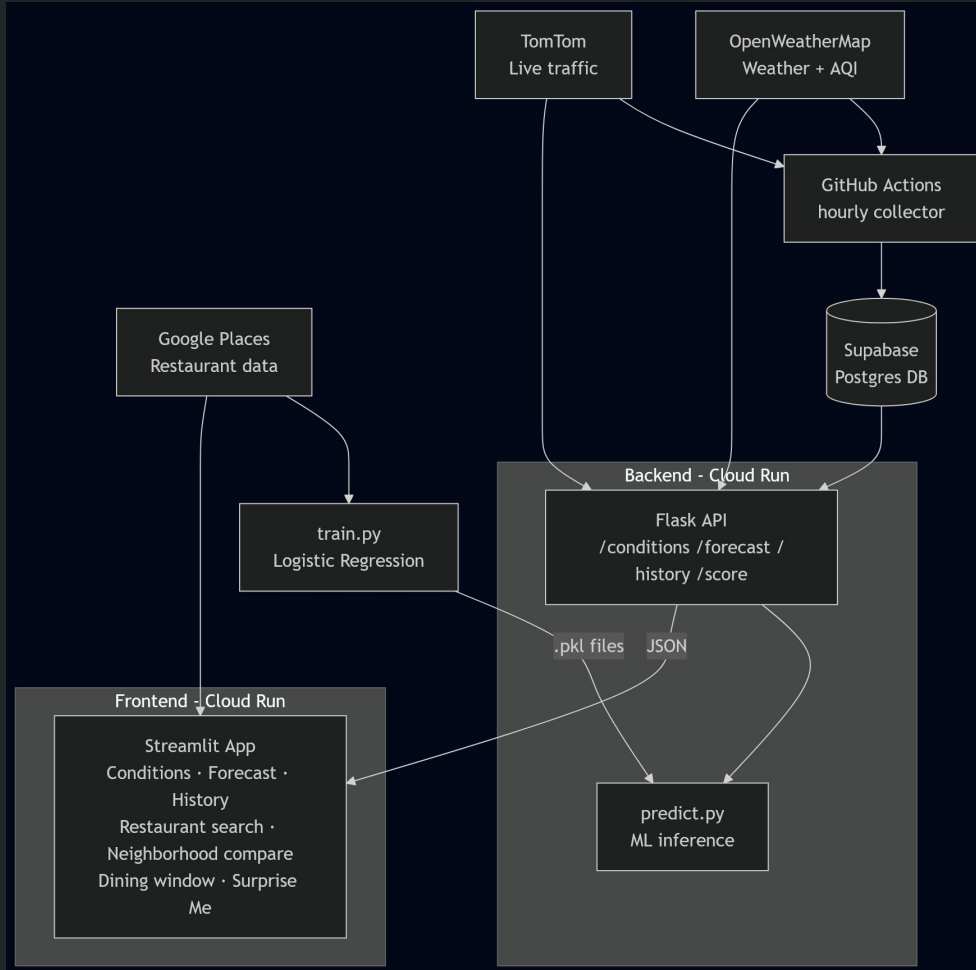
Next 6-9 Hours of weather + AQI.

7-Day Historical Trends that give temperature, AQI, and congestion

Live neighborhood comparison in a conditions table

Restaurant Search and Top Restaurant Browse that gives an ML Score, the Google Rating, and a Combined Score

Surprise Me Function that picks a random neighborhood and restaurant. Stores last 5 so you can compare.



Demoing the Application: KBBQ in Rowland Heights

Settings

Neighborhood

Rowland Heights

Preference Weights

Adjust what matters most to you

Weather



Air Quality



Traffic



Surprise Me!

Should I Dine Out in LA Today?

Real-time conditions and ML restaurant scoring for the Greater Los Angeles area

Rowland Heights

Temp

66.96F

↑ Feels 66.74F

Humidity

72%

Wind

11.5 mph

AQI

Fair

↑ PM2.5: 8.93 ug/m3

Traffic

Moderate

↑ 38% congestion

Sky: Clear Sky | Rain last hour: 0.0in | Conditions score: 75.7/100 | Conditions as of 08:09 PM PDT

Next Few Hours

Temperature Forecast (F)



AQI Forecast

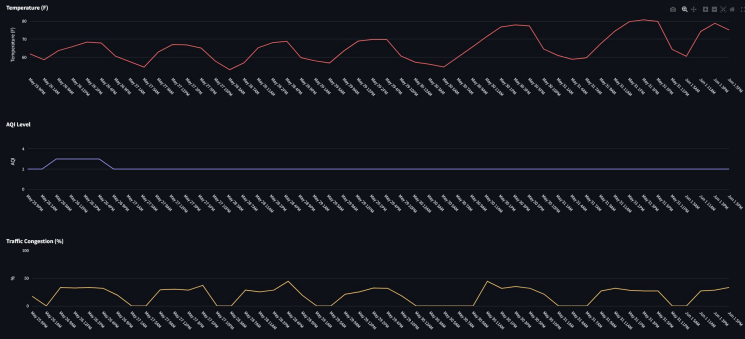
Jun 1 11PM — Fair (AQI 2)

Jun 2 2AM — Fair (AQI 2)

Jun 2 5AM — Fair (AQI 2)

AQI scale: 1=Good, 2=Fair, 3=Moderate, 4=Poor, 5=Very Poor

7-Day Historical Trends



Dining Window Planner

Find the best time to go based on forecast conditions

Time (LA)	Temp (F)	AQI	Rain (inch)	Wind (mph)	Conditions Score
Jun 1 11PM	66.96000	Fair	0.00000	11.50000	75.70000
Jun 2 2AM	64.00000	Fair	0.00000	11.50000	77.80000
Jun 2 5AM	61.50000	Fair	0.00000	11.50000	77.50000

Best time to go: Jun 1 11PM (conditions score: 75.7/100) — highlighted in green above

Traffic uses current live reading — forecast traffic not available on TomTom free tier

Neighborhood Comparison

Compare current conditions across multiple neighborhoods side by side

Select neighborhoods to compare

Rowland Heights 5 Rowland Heights 4 Pasadena 4

Neighborhood	Temp (F)	Feels Like (F)	Humidity (%)	Wind (mph)	AQI	Traffic	Congestion (%)	Conditions Score
Rowland Heights	66.96	66.74	72	11.5	Fair	Moderate	38.51	76.3/100
Koreatown	65.28	65.14	71	9.37	Fair	Moderate	34.52	77.3/100
Pasadena	67.08	66.72	69	4.16	Fair	Light	0	96.6/100

Find a Restaurant

[Search by Name](#) [Browse Neighborhood](#)

Restaurant name

Thirsty Cow

Select a restaurant

Thirsty Cow Korean BBQ

Address: 17500 Castleton St, City of Industry, CA 91748, USA

[GO] Go for it! — Thirsty Cow Korean BBQ in Rowland Heights

Restaurant Tier

Worth the trip

ML Confidence

66%

Conditions Score

76.2/100

Rating

4.5 stars

↑ 1,367 reviews

See the Probability Breakdown

Decent: 27%

Must go: 6%

Skip it: 1%

Worth the trip: 66%

Moderate traffic in Rowland Heights currently (37% congestion).

Find a Restaurant

Search by Name [Browse Neighborhood](#)

Browse top restaurants in this neighborhood

Select a restaurant

Chubby Cattle BBQ | Rowland Heights

Address: 1388 Fullerton Rd, Rowland Heights, CA 91748, USA

[GO] Go for it! — Chubby Cattle BBQ | Rowland Heights in Rowland Heights

Restaurant Tier

Must go

ML Confidence

94%

Conditions Score

76.2/100

Rating

4.9 stars

↑ 11,196 reviews

See the Probability Breakdown

Decent: 0%

Must go: 94%

Skip it: 0%

Worth the trip: 6%

Moderate traffic in Rowland Heights currently (37% congestion).

Find a Restaurant

Search by Name Browse Neighborhood

Restaurant name

kbbq

Select a restaurant

88Q Korean BBQ

88Q Korean BBQ

Baekjeong Rowland Heights

Thirsty Cow Korean BBQ

Chubby Cattle BBQ | Rowland Heights

Yangmani K.BBQ Rowland Heights

Chamsutgol BBQ

BBQ STOP AYCE - Diamond Bar

Gui Gui 9292 Korean BBQ (AYCE) - Rowland Heights

Surprise Me!

Taking you to West Hollywood!

Recent Surprise Picks

Roots Indian Bistro — West Hollywood

Restaurant Tier

Must go

ML Confidence

94%

Conditions Score

80.8/100

Rating

4.9 stars

↑ 13,308 reviews

Address: 8565 Melrose Ave, West Hollywood, CA 90069, USA

Urban Plates — Torrance

Mayura Indian Restaurant — Culver City

Girl & the Goat Los Angeles — Downtown LA

555 East American Steakhouse — Long Beach

The Solution

Technical Challenges

Data Collection (Lack of Public APIs + Occasional Cronjob Failures)

AI Assistant Use

READMEs, Debugging Code, Streamlit UX Improvements, Flask API

Potential Improvements

Retrain Model, Compare Searches, Expand Coverage, Save Favorites

Thank you. Fin